



National Disaster
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BRD- AapdaMitra & MIS

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Table of Contents

1. Introduction	3
1.1. Purpose	3
1.2. Background	4
1.3. Need for Revamp and Unification	5
1.4. Vision for the Unified Aapda Mitra Platform	5
2. Objectives	6
2.1. Business Objectives	6
2.2. Functional Objectives	6
2.3. Technical Objectives	7
2.4. UI/UX Objectives	8
2.5. Reporting & Analytics Objectives	8
2.6. Operational Objectives	8
3. Scope of Work	8
3.1. In-Scope	11
3.2. Out of Scope.	12
4. Stakeholders	12
5. Functional Requirements	14
5.1. Volunteer Onboarding	14
5.2. VOLUNTEER VERIFICATION MODULE	15
5.3. TRAINING MANAGEMENT MODULE	15
5.4. TRAINING MANAGEMENT MODULE	16
5.5. ERK KIT & INSURANCE MODULE	16
5.6. DEPLOYMENT & FIELD OPERATIONS MODULE	16
5.7. MONTHLY REPORTING MODULE	16
5.8. FEEDBACK MODULE	17
5.9. DASHBOARDS & ANALYTICS MODULE	17
5.10. DASHBOARDS & ANALYTICS MODULE	17
5.11. USER MANAGEMENT & ACCESS CONTROL	17
6. Non-Functional Requirements	18
6.1. Performance Requirement	18
6.2. Availability & Reliability Requirements	18



	18
6.3. Security Requirement	19
6.4. Scalability Requirements	19
6.5. Usability & Accessibility Requirements	19
6.6. Interoperability Requirements	20
6.7. Maintainability & Support Requirements	20
6.8. Data Quality & Integrity Requirements	21
7. Business Rules / Business Logic	21
7.1. Registration & Eligibility Rules	22
7.2. Verification Rules	22
7.3. Training Rules	22
7.4. Certification Rules	23
7.5. ERK Kit & Insurance Rules	23
7.6. Deployment & Geo-Tracking Rules	24
7.7. Monthly Reporting Rules	24
7.8. Feedback Rules	24
7.9. Dashboard Rules	24
7.9. Data Integrity & Governance Rules	25
8. Portal Work Flow	25
8.1. Workflow 1 — Volunteer Registration & Onboarding	25
8.2. Workflow 2 — Youth Organisation Affiliation	26
8.3. Workflow 3 — Volunteer Verification & Approval	26
8.4. Workflow 4 — Training Allocation	26
8.5. Workflow 5 — Attendance & Evidence Capture	26
8.6. Workflow 6 — Certification (Physical + DigiLocker)	27
8.7. Workflow 7 — ERK Kit & Insurance Issuance	27
8.8. Workflow 8 — Deployment & Field Operations	28
8.9. Workflow 9 — Monthly Reporting & Compliance	28
8.10. Workflow 10 — Feedback Management	28
8.11. Workflow 11 - Public Dashboard Updates	29
8.12. Workflow 12 — MIS Dashboards (DDMA, SDMA, NDMA)	29
12. Key Abbreviation	33

1. Introduction

1.1. Purpose

This Business Requirements Document (BRD) defines the complete functional, non-functional, business, integration, and reporting requirements for the **Unified Aapda Mitra Volunteer Management Platform** to be developed for the National Disaster Management Authority (NDMA).

The objective of this document is to ensure a shared and consistent understanding among NDMA, SDMA, DDMA, Training Institutes, Youth Organisations, and NeGD on the processes, workflows, services, roles, and system capabilities required for the modernization of the existing Aapda Mitra ecosystem.

This BRD will serve as the primary reference for design, development, validation, implementation, and future enhancement of the unified Aapda Mitra system.

1.2. Background

The Aapda Mitra initiative was originally launched to build a trained cadre of community volunteers capable of responding to disasters at the district and state levels. Over the years, two distinct digital systems evolved:

1. **Aapda Mitra Public Portal** – where volunteers viewed scheme details and registered.
2. **Aapda Mitra MIS (Internal Portal)** – used by DDMA, SDMA, and NDMA for training workflows, attendance monitoring, ERK distribution, and reporting.

With the rollout of the **Upscaling of Aapda Mitra Scheme (UAMS)** and the recently introduced **Yuva Aapda Mitra Scheme (YAMS)**, NDMA now requires a unified national platform that can:

- Manage volunteers from **public applicants** and **youth organisations** (NCC, NSS, NYKS, BS&G, MyBharat, NCSC).
- Digitize field-level evidence and monitoring processes.
- Strengthen deployment readiness during disaster events.
- Provide real-time dashboards for officials and public transparency.

The current systems create duplication, inconsistent records, manual dependencies, and limited visibility across stakeholders. A single consolidated platform is essential to support scalability, accuracy, and efficient coordination.

1.3. Need for Revamp and Unification

Key factors driving the modernization of the Aapda Mitra ecosystem include:

1. Fragmented Systems

Two independent systems (public portal + MIS) result in separated workflows and duplicated data. The unified portal will integrate both into one seamless platform.

2. Mandatory Volunteer Data Expansion

NDMA requires a **30-field volunteer profile**, including mandatory fields such as **MyBharat ID**, Aadhaar, eligibility criteria, and organisational affiliation.

This cannot be handled effectively by the existing system.

3. New Scheme-Specific Requirements (YAMS)

The YAMS 2024–26 scheme introduces:

- Digitally submitted youth organisation volunteer lists
- New training models and syllabi
- Updated ERK kit and insurance workflows
- Master Trainer (ToT) program
- API-based certificate issuance (DigiLocker) for new scheme volunteers

These require dedicated system support.

4. Evidence-Based Governance

Updated NDMA guidelines mandate:

- QR-based attendance
- Geo-tagged photo/video evidence
- NDMA Super Admin approval flow

The existing MIS does not support this rigor.

5. Integration with External Systems

The unified platform must integrate with:

- **Aadhaar Authentication**
- **DigiLocker**
- **Youth Organisation APIs**
- **NDMA-Internal services for geo-location tracking**

- **SMS/Email/WhatsApp gateways**

The present system does not support these integrations.

6. Public and Government Dashboard Requirements

NDMA requires:

- A **public-facing dashboard** to showcase national scheme performance
- **Role-based MIS dashboards** for DDMA, SDMA, and NDMA officials for real-time monitoring, compliance, and analytics

1.4. Vision for the Unified Aapda Mitra Platform

The vision of the revamped system is to create a **national, scalable, integrated volunteer management ecosystem** that:

- Serves as the single source of truth for all Aapda Mitra volunteer records
- Standardizes processes across all states, districts, and youth organisations
- Enables transparent and digital monitoring of the training lifecycle
- Ensures readiness of volunteers for deployment through structured tracking
- Supports decision-making using dashboards, analytics, and geo-intelligence
- Integrates seamlessly with government digital platforms
- Provides a mobile-first experience for volunteers and officials
- Upholds compliance with GIGW, accessibility, DPDP Act, and security standards

2. Objectives

The unified Aapda Mitra platform aims to modernize and standardize volunteer lifecycle management across the Upscaling Aapda Mitra Scheme (UAMS) and the Yuva Aapda Mitra Scheme (YAMS), ensuring transparency, operational efficiency, and scalable disaster-response readiness nationwide.

2.1. Business Objectives

- **Create a Unified National Platform**
Consolidate the current Aapda Mitra public portal and MIS system into a single, integrated volunteer management ecosystem supporting both UAMS (legacy scheme) and YAMS (youth-driven scheme).
- **Strengthen Volunteer Lifecycle Governance**
Digitize and standardize all key processes—registration, verification, training, attendance, ERK & insurance provisioning, deployment, monthly reporting, and feedback—across all states and districts.
- **Enable Youth Organisation Participation**
Support digital volunteer onboarding via NCC, NSS, BS&G, NYKS, MyBharat, and NCSC through API-based data intake and structured verification workflows.
- **Improve Training Quality and Transparency**
Ensure accurate attendance (QR + geo-tag), photo/video evidence capture by nodal officers, and NDMA Super Admin approval for all training evidence.

- **Enhance Deployment Readiness**
Enable GPS/geo-tracking via Sachet App, readiness scoring, and real-time operational situational awareness for NDMA, SDMA, and DDMA.
- **Increase Public Transparency**
Provide a public-facing dashboard on the AapdaMitra portal showcasing national progress, coverage, training outcomes, and scheme impact.
- **Support Data-Driven Decision Making**
Equip NDMA with advanced dashboards, analytics, KPIs, and predictive insights enabling resource planning and performance monitoring.

2.2. Functional Objectives

- **Volunteer Registration & Profiling**
 - Capture a **30-field profile** for each volunteer, with **MyBharat ID as a mandatory field**.
 - Enable Aadhaar-based identity verification.
 - Allow youth organisation data ingestion through API/file uploads.
- **Automated Eligibility Determination**
 - Replace manual SDMA approval with auto-eligibility logic based on scheme criteria and youth organisation validation.
- **Training Lifecycle Management**
 - Manage batch creation, scheduling, volunteer allocation, and training institution mapping.
 - Enable QR-based attendance and geo-tagged evidence capture by **District Nodal Officers**.
 - Ensure NDMA Super Admin reviews and approves evidence submissions.
- **Certification Workflow**
 - Support **physical certificates** for legacy (UAMS) volunteers.
 - Enable **API-based DigiLocker certificates** for new YAMS volunteers.
- **ERK Kit & Insurance Tracking**
 - Maintain ERK kit issuance records and insurance activation details.
 - Integrate ERK and insurance feedback within the overall volunteer feedback module.
- **Deployment & Field Tracking**
 - Integrate Sachet App for:
 - Live GPS trail of volunteers (new scheme)
 - Location mapping for older scheme volunteers via Excel upload
 - Enable task assignment and real-time deployment updates.
- **Monthly Reporting & Compliance**
 - Capture monthly readiness updates.
 - Trigger reminders, delinquency alerts, and escalation workflows.
- **Feedback Management**
 - Collect structured feedback covering training, ERK, insurance, and volunteer satisfaction.
- **Role-Based Access & Dashboards**
 - Provide custom dashboards for NDMA, SDMA, DDMA, Training Institutes, and Volunteers.
 - Provide a **public-facing, non-PII dashboard** on the AapdaMitra portal.
- **Document & Policy Management**
 - Allow NDMA to publish notifications, guidelines, SOPs, and training materials via CMS.

2.3. Technical Objectives

- **Modular Microservices Architecture**
 - Build a scalable, interoperable backend that supports continuous enhancement, high concurrency, and national rollout.
- **Standards-Based Digital Integrations**

Integrate securely with:

 - Aadhaar Authentication API
 - DigiLocker (New Scheme Certificates)
 - Internal APIs
 - Youth Organisation APIs
 - SMS/Email/WhatsApp Gateways
 - Maps/GIS Services
- **Unified Master Data Layer**
 - Consolidate volunteer, training, evidence, ERK, insurance, deployment, and feedback datasets into a common data model.
- **Robust Security Framework**
 - Implement RBAC, MFA for officials, encryption-at-rest, encryption-in-transit, audit trails, and DPDP Act compliance.
- **GIGW & Accessibility Compliance**
 - Ensure the portal and dashboards are GIGW compliant.

2.4. UI/UX Objectives

- Deliver an intuitive, mobile-first UI for volunteers and officers.
- Provide simplified forms for 30-field volunteer entry.
- Ensure dashboards have accessible drill-down paths for NDMA, SDMA, DDMA.
- Make the public-facing dashboard visually appealing, simple, and multilingual.
- Enable consistency of design and experience across web and mobile channels.

2.5. Reporting & Analytics Objectives

- Offer a **public dashboard** highlighting scheme reach, volunteer strength, training progress, and state-level metrics.
- Provide **internal MIS dashboards** with role-based data views:
 - NDMA: National KPIs, inspection insights, geo-intelligence
 - SDMA: State readiness, training compliance, ERK/insurance status
 - DDMA: District training monitoring, volunteer readiness
- Deliver predictive models for:
 - Volunteer readiness index
 - District performance benchmarking
 - Training demand planning
 - High-risk zone mapping

- Enable exportable reports for state and national-level review.

2.6. Operational Objectives

- Reduce manual processes across DDMA, SDMA, and Youth Organisations.
- Improve training oversight through centralized evidence review and structured workflows.
- Increase readiness through consistent monthly reporting mechanisms.
- Ensure faster mobilization of volunteers during disasters through real-time monitoring.
- Maintain a continuously updated national volunteer registry.

3. Scope of Work

The Scope of Work (SoW) for the Unified Aapda Mitra Volunteer Management Platform defines all functional, technical, operational, analytical, and integration capabilities required to support the end-to-end volunteer lifecycle across the Upscaling Aapda Mitra Scheme (UAMS) and Yuva Aapda Mitra Scheme (YAMS).

The unified system will replace the existing Aapda Mitra Public Portal and MIS system with a consolidated, scalable, and modern digital platform.

3.1. In-Scope

The following functionalities and capabilities are included within the scope of this initiative:

A. Volunteer Lifecycle Management

1. Volunteer Registration & Onboarding

- Support public registration and youth organisation–driven registrations.
- Introduce a **30-field volunteer profile** including **mandatory MyBharat ID**, Aadhaar, demographic details, and eligibility parameters.
- Enable profile completion by DDMA/State Nodal Officers.

2. Volunteer Verification

- Aadhaar authentication integration.
- Youth Organisation affiliation validation.
- **Auto-eligibility determination** based on scheme criteria (no manual SDMA approval).

3. Training Management

- Training institute onboarding & batch creation.
- Volunteer-to-batch allocation by DDMA/SDMA.
- Geo-tagged photo/video evidence by nodal officers.
- NDMA Super Admin review and approval of evidence.

4. Certification Workflow

- Physical certificate management for UAMS (Old Scheme).
- API-based DigiLocker certificate generation for YAMS (New Scheme).

5. ERK Kit & Insurance



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- ERK kit distribution tracking.
- Insurance enrollment and status management.
- ERK & Insurance feedback integrated into volunteer feedback form.

6. **Deployment & Operational Readiness**

- Internal services integration for real-time GPS tracking (new scheme).
- Excel-based location updates for legacy volunteers.
- Task assignment workflows for DDMA/SDMA/NDMA during emergencies.
- Real-time deployment status updates from volunteers.

7. **Monthly Reporting**

- Capture volunteer readiness, activity logs, and willingness to deploy.
- Auto-reminders and delinquency escalation mechanisms.

8. **Feedback Management**

- Capture feedback on training experience, ERK kit, insurance, and overall satisfaction.

B. Youth Organisation Enablement

1. API-based or file-based volunteer data ingestion from:

- NCC
- NSS
- NYKS
- Bharat Scouts & Guides (BS&G)
- MyBharat
- NCSC

2. Validation of affiliation by youth organisations.

3. Monitoring dashboards for youth organisations.

C. Role-Based Interfaces & Dashboards

Public-Facing Dashboard (AapdaMitra Website)

- Scheme overview & metrics
- National volunteer coverage
- Training statistics
- Youth organisation contributions
- State/district coverage heatmap
- Success stories, gallery, and announcements

Internal MIS Dashboards

1. **DDMA Dashboard**

- District-level volunteer status
- Batch/attendance/evidence monitoring
- Deployment readiness & compliance

2. SDMA Dashboard

- State-level volunteer lifecycle indicators
- Training capacity & progress
- ERK/insurance status
- District performance analytics

3. NDMA Dashboard

- National performance KPIs
- Geo-intelligence
- Evidence compliance review
- High-level insights & predictive analytics

4. Training Institute Dashboard

- Batch rosters
- Training calendar

D. Data Management & Migration

1. Migrating existing UAMS volunteer data into the unified system.
2. Normalizing legacy records to comply with the new 30-field data model.
3. Backward compatibility for volunteers missing MyBharat ID.
4. Upload workflows for old scheme location mapping (Excel-based).

E. System Integrations

1. **Aadhaar Authentication**
 - Identity verification using UIDAI-approved mechanisms.
2. **DigiLocker Integration**
 - Certificate generation for YAMS volunteers.
3. **Internal Services Integration**
 - Real-time volunteer tracking during emergencies.
4. **Communication Services**
 - SMS, Email, WhatsApp API for notifications and alerts.
5. **GIS / Maps API**
 - Geo-mapping for deployments and training locations.
6. **Youth Organisation APIs**
 - Secure intake of volunteer datasets.
7. **Content Management System (CMS)**
 - NDMA to publish guidelines, policy documents, manuals, and announcements.

F. Evidence Management

1. QR-based attendance proof.
2. Geo-tagged photos/videos uploaded by nodal officers.
3. Centralized evidence repository with NDMA approval workflow.

4. Compliance tracking dashboards.

G. Security & Access Control

1. Role-based access for Volunteers, Youth Orgs, DDMA, SDMA, NDMA, and Institutes.
2. MFA for official users.
3. Audit trails for all actions.
4. Encryption of data at rest and in transit.
5. DPDP Act and GIGW 3.0 compliance.

H. Reporting & Analytics

1. State, district, and national MIS reports.
2. Drill-down analytics for training, volunteer performance, ERK, insurance, deployment, and feedback.
3. Exportable reports (PDF, Excel, CSV).
4. Predictive analytics for readiness and training demand.

3.2. Out of Scope.

To avoid ambiguity, the following are explicitly out of scope:

1. **Physical execution of training programs** (handled offline by training institutions).
2. **Procurement & physical distribution** of ERK kits (system only tracks records).
3. **Insurance underwriting or claim settlement** workflows.
4. **Hardware procurement** for states or districts.
5. **Operational management of field deployments** (system supports tracking and reporting only).
6. **Mobile hardware for volunteers** (BYOD model assumed).
7. **NDRF/SDRF operational decision-making processes** outside digital reporting.

4. Stakeholders

The unified Aapda Mitra platform engages a diverse set of stakeholders across national, state, district, training, and youth-organisation ecosystems. Each stakeholder group plays a unique role in the volunteer lifecycle, from data submission and verification to training, monitoring, deployment, and reporting.

This section describes all stakeholders, their roles, access rights, and responsibilities within the system.

Stakeholder	Role Category	Description	Key Responsibilities in System
Volunteer (UAMS/YAMS)	Primary End User	Individual participating in Aapda Mitra training, deployment, and readiness programs.	Register/consent, provide documents, receive allocations, attend training, update deployment status, submit monthly readiness, provide feedback.



Youth Organizations (NCC, NSS, BS&G, NYKS, NCSC, MyBharat)	External Data Providers	Provide curated volunteer lists under YAMS and validate affiliation.	Submit volunteer datasets (API/file), confirm affiliation, support scheme outreach, track volunteer status.
District Nodal Officer (DDMA)	District Administrator	Performs volunteer onboarding, profile completion, verification, batch allocation, training monitoring, and deployment coordination.	Complete 30-field profile, validate volunteer eligibility, allocate volunteers to training, upload training evidence, monitor deployment updates, escalate issues.
State Authority (SDMA)	State Administrator	Oversees scheme execution at state level, reviews district performance, handles insurance/ERK workflows, and monitors readiness.	Auto-eligibility oversight, institute onboarding, ERK & insurance updates, compliance review, state-level dashboard monitoring.
NDMA	Authority	Governing body responsible for national coordination, policy management, approval workflows, and national dashboards.	Evidence approval, policy publishing, YAMS/UAMS governance, monitoring analytics, conducting field inspections, overseeing scheme KPIs.
Training Institute & Trainers	Training Execution Entities	Physical sites delivering 7-day training programs.	Provide training infrastructure, conduct sessions, coordinate with nodal officers for verification and evidence capture. (They do not upload evidence digitally.)
Super Admin (NDMA)	System Oversight	Highest-level administrative access for data governance, evidence approval, scheme configuration, and user management.	Approve evidence, manage user roles, configure scheme settings, oversee system integrity.



System Administrator (NeGD/NDMA ICT)	Technical Administrator	Responsible for platform configuration, monitoring, system support, and managing backend services.	User provisioning, access management, CMS configuration, integration monitoring, technical troubleshooting.
Public Users (Citizens/Researchers)	Information Consumers	General public viewing scheme data through the public dashboard.	Access public-facing metrics, follow scheme details, no login or personal data access.

5. Functional Requirements

Functional Requirements define what the system **must provide** to meet NDMA's operational, training, governance, and monitoring needs. Each requirement has been assigned a unique Functional Requirement ID (FR-ID) grouped according to process modules.

5.1. Volunteer Onboarding

1) FR-REG-001: Registration Modes

- The system shall support registration modes:
 - a) Youth Organisation-driven mode (bulk/API submission)

2) FR-REG-002: 30-Field Volunteer Profile Capture

- The system shall allow DDMA Nodal Officers to complete a 30-field mandatory volunteer profile, including personal, demographic, address, and training preference details.

3) FR-REG-003: Mandatory MyBharat ID

- The system shall enforce MyBharat ID as a mandatory field for all volunteers (UAMS & YAMS).

4) FR-REG-004: Aadhaar Authentication

- The system shall integrate Aadhaar authentication to validate volunteer identity.

5) FR-REG-005: Auto-Duplication Check

- The system shall detect duplicate volunteers using Aadhaar + mobile + MyBharat ID.

6) FR-REG-006: Volunteer Consent Capture

- The system shall record volunteer consent for data usage, youth organisation data intake, and YAMS enrollment.

7) FR-REG-007: Youth Organisation Data Intake

- The system shall allow Youth Organisations to upload or submit volunteer lists via APIs or secure upload.

8) FR-REG-008: Status Notification

- The system shall notify volunteers about registration, profiling, and verification progress via SMS/Email/WhatsApp.



5.2. VOLUNTEER VERIFICATION MODULE

1) **FR-VER-001: DDMA Profile Validation**

- The system shall allow DDMA Nodal Officers to validate volunteer eligibility and update verification status.

2) **FR-VER-002: Auto-Eligibility Logic**

- The system shall automatically determine eligibility based on NDMA-defined rules (AGE, PHYSICAL FITNESS, YOUTH ORG membership, etc.).

3) **FR-VER-003: Youth Organisation Affiliation Verification**

- The system shall allow Youth Organisations to confirm or reject volunteer affiliation.

4) **FR-VER-004: Verification Dashboard**

- The system shall provide DDMA/SDMA dashboards showing:
 - Pending verifications
 - Verified volunteers
 - Rejected volunteers
 - Data mismatches

5) **FR-VER-005: Verification Audit Trail**

- The system shall maintain logs of all verifications performed.

5.3. TRAINING MANAGEMENT MODULE

1) **FR-TRN-001: Training Institute Onboarding**

- The system shall allow SDMA to onboard training institutes with details of capacity, infrastructure, and location.

2) **FR-TRN-002: Batch Creation**

- The system shall allow DDMA/SDMA to create training batches with start/end dates, capacity, trainers, and district mapping.

3) **FR-TRN-003: Volunteer Allocation**

- The system shall allow DDMA to allocate verified volunteers to available batches.

4) **FR-TRN-004: Training Notifications**

- The system shall send notifications to volunteers upon batch allocation.

5) **FR-TRN-005: Attendance Capture (QR-Based)**

- The system shall support QR scanning by the Nodal Officer for daily attendance capture.

6) **FR-TRN-006: Geo-Tagged Attendance Evidence**

- The system shall allow Nodal Officers to upload geo-tagged photos/videos as evidence for each training day.

7) **FR-TRN-007: Offline Attendance Support**

- The system shall allow attendance to be recorded offline and synced later when network becomes available.

8) **FR-TRN-008: Evidence Approval Workflow**

- The system shall route all evidence to **NDMA Super Admin** for approval.

9) **FR-TRN-009: Attendance Dashboard**

- The system shall show attendance summary, absents, compliance %, and evidence status to DDMA/SDMA/NDMA.

5.4. TRAINING MANAGEMENT MODULE

1) FR-CERT-001: Eligibility for Certificate

The system shall auto-compute eligibility based on:

- Attendance %
 - Evidence approval
 - NDMA defined rules
- ### 2) FR-CERT-002: Physical Certificate (Old Scheme)
- The system shall allow certificate status tracking for UAMS (physical certificate issuance).
- ### 3) FR-CERT-003: DigiLocker Integration (New Scheme)
- The system shall integrate with DigiLocker for issuance of digital certificates for YAMS volunteers.
- ### 4) FR-CERT-004: Certificate Sharing
- The system shall allow volunteers to download, view, and share their certificate.

5.5. ERK KIT & INSURANCE MODULE

1) FR-ERK-001: ERK Kit Distribution Tracking

- The system shall track the ERK kit distribution for each volunteer.

2) FR-ERK-002: Insurance Enrollment

- The system shall record insurance policy number, activation date, and expiry date.

3) FR-ERK-003: Volunteer Acknowledgment

- The system shall allow volunteers to acknowledge ERK kit & insurance receipt via mobile.

4) FR-ERK-004: ERK/Insurance Feedback

- The system shall capture kit and insurance feedback within the unified feedback form.

5.6. DEPLOYMENT & FIELD OPERATIONS MODULE

1) FR-DEP-001: Task Assignment

- The system shall allow NDMA/SDMA/DDMA to assign deployment tasks during emergencies.

2) FR-DEP-002: Sachet App Integration

- The system shall integrate the Sachet App for real-time location tracking of new YAMS volunteers.

3) FR-DEP-003: Legacy Volunteer Tracking Upload

- The system shall allow Excel-based location mapping for UAMS volunteers.

4) FR-DEP-004: Live Field Activity Updates

The system shall allow volunteers to submit deployment updates:

- Location
- Task progress

- Incident notes
- 5) **FR-DEP-005: Deployment Dashboard**
- The system shall provide NDMA with real-time deployment heatmaps and volunteer readiness index.

5.7. MONTHLY REPORTING MODULE

- 1) **FR-MON-001: Monthly Readiness Submission**
- The system shall allow volunteers to submit their monthly readiness and activity status.
- 2) **FR-MON-002: Automated Reminders**
- The system shall trigger reminders for non-compliant volunteers.
- 3) **FR-MON-003: Delinquency Escalation**
- After repeated non-compliance, the system shall escalate to DDMA/SDMA dashboards.

5.8. FEEDBACK MODULE

- 1) **FR-FDB-001: Unified Feedback Form**

The system shall capture feedback for:

- Training
 - ERK kit
 - Insurance
 - Overall experience
- 2) **FR-FDB-002: Feedback Analytics**
- The system shall feed feedback data to analytics for insights.

5.9. DASHBOARDS & ANALYTICS MODULE

- 1) **FR-DASH-001: Public Dashboard**

The system shall provide a public dashboard displaying non-PII scheme metrics:

- Total volunteers
 - Total trained
 - State/district coverage
 - Training status
 - Youth organisation participation
- 2) **FR-DASH-002: DDMA Dashboard**
- Shows: batch progress, attendance, evidence status, volunteer readiness.
- 3) **FR-DASH-003: SDMA Dashboard**
- Shows: state readiness, ERK/insurance, district performance.
- 4) **FR-DASH-004: NDMA Dashboard**
- Shows: national KPIs, geo-intelligence, deployment monitoring, scheme analytics.
- 5) **FR-DASH-005: Youth Organisation Dashboard**

- Shows volunteer list status, training progress, and affiliation confirmations.

5.10. DASHBOARDS & ANALYTICS MODULE

1) FR-CMS-001: Publish Guidelines & SOPs

- NDMA shall be able to publish scheme documents, training materials, and updates.

2) FR-CMS-002: Multi-Lingual Support

- The system shall support content publishing in multiple languages.

5.11. USER MANAGEMENT & ACCESS CONTROL

1) FR-USER-001: Role-Based Access Control (RBAC)

- The system shall assign roles: Volunteer, Youth Org, DDMA, SDMA, NDMA, Super Admin, Technical Admin.

2) FR-USER-002: Multi-Factor Authentication

- The system shall enforce MFA for official users.

3) FR-USER-003: Audit Logging

- The system shall record all critical user actions.

6. Non-Functional Requirements

Non-Functional Requirements describe the quality attributes, constraints, and operational expectations of the unified Aapda Mitra platform. These requirements ensure that the system is reliable, secure, scalable, and usable for nationwide deployment across NDMA, SDMA, DDMA, Training Institutes, and Youth Organisations.

6.1. Performance Requirements

1) NFR-PERF-001: Response Time

- All key application pages (Portal + MIS) shall load within **2–3 seconds** under normal load.
- APIs shall respond within **<1 second** for 90% of requests.

2) NFR-PERF-002: Attendance Upload Performance

- Geo-tagged photo uploads must complete within **≤ 5 seconds** under 4G/5G network.
- Offline evidence sync must occur automatically when network is restored.

3) NFR-PERF-003: Dashboard Rendering

- Public and internal dashboards must load and render in **< 4 seconds** even with large datasets.

4) NFR-PERF-004: Scalability for Volunteer Base

- The system must handle up to **1 million+ volunteers** with concurrent district-level operations.

6.2. Availability & Reliability Requirements

1. NFR-AVAIL-001: Uptime

- The system must maintain **99.9% uptime** excluding planned maintenance.

2. NFR-AVAIL-002: Disaster Recovery (DR)

- RPO (Recovery Point Objective): **15 minutes**
- RTO (Recovery Time Objective): **1 hour**
- 3. **NFR-AVAIL-003: Backup Requirements**
 - Full database backup every 24 hours; incremental backups hourly.
 - Evidence repository backups every 12 hours.

6.3. Security Requirements

1. **NFR-SEC-001: Aadhaar Data Compliance**
 - Aadhaar verification must follow UIDAI guidelines.
 - Aadhaar numbers must **not be stored in plain text**.
2. **NFR-SEC-002: Encryption**
 - All data in transit must use **TLS 1.2+**.
 - All data at rest must use **AES-256 encryption**.
3. **NFR-SEC-003: Role-Based Access Control (RBAC)**
 - Fine-grained access control must be enforced for:
Volunteers, Youth Organisations, DDMA, SDMA, NDMA, Super Admin, Technical Admin.
4. **NFR-SEC-004: Multi-Factor Authentication**
 - MFA mandatory for NDMA, SDMA, DDMA, Training Institute, Super Admin roles.
5. **NFR-SEC-005: Audit Trails**
 - Every critical action (verification, evidence approval, data modification, batch assignment, certificate issuance) must be logged with:
 - User ID
 - Timestamp
 - IP Address
 - Action Summary
6. **NFR-SEC-006: DPDP Act Compliance**
 - Data minimization, purpose limitation, consent capture, and retention policies must adhere to the Digital Personal Data Protection (DPDP)

6.4. Scalability Requirements

1. **NFR-SCALE-001: Horizontal & Vertical Scaling**
 - Backend microservices must support horizontal scaling during peak load (training cycles, deployment cycles, monthly reporting, evidence spikes).
2. **NFR-SCALE-002: File Storage Scalability**
 - Evidence storage (photos/videos) must be able to scale to **multi-terabyte** size.
3. **NFR-SCALE-003: Youth Organisation Integrations**
 - Platform must support concurrent data ingestion from **all six organisations** (NCC, NSS, NYKS, BS&G, MyBharat, NCSC)

6.5. Usability & Accessibility Requirements

1. NFR-USE-001: GIGW 3.0 Compliance

- The unified portal must comply with GIGW 3.0 standards, including accessibility, multilingual content, and responsive layouts.

2. NFR-USE-002: WCAG 2.1 AA Compliance

- Ensure accessible navigation for low-vision users, screen reader compatibility, and color contrast standards.

3. NFR-USE-003: Mobile-First Design

- UI must be optimized for mobile (3G/4G/5G) to support field officer usage.

4. NFR-USE-004: Offline Mode

- Offline attendance capture (QR + photos) must be supported for regions with poor connectivity.

5. NFR-USE-005: Low-Bandwidth Optimization

- Dashboard and evidence screens must be usable on networks as low as 512 kbps.

6.6. Interoperability Requirements

1) NFR-INT-001: Aadhaar API Integration

- Must conform to UIDAI API and security standards.

2) NFR-INT-002: DigiLocker API Integration

- Certificate generation for new scheme (YAMS) must follow DigiLocker issuing partner guidelines.

3) NFR-INT-003: Sachet App Integration

- Real-time tracking for YAMS volunteers must sync via Sachet APIs.
- Support Excel upload for old scheme tracking.

4) NFR-INT-004: Youth Organisation API / Bulk Data Intake

- NCC, NSS, BS&G, NYKS, MyBharat, NCSC must be supported through API or secure upload workflows.

5) NFR-INT-005: Notification Gateways

- SMS, Email, and WhatsApp gateways must integrate using configurable adapters.

6) NFR-INT-006: GIS / Maps Services

- The platform must integrate with geospatial APIs to support evidence mapping and deployment visualization.

6.7. Maintainability & Support Requirements

1) NFR-MAIN-001: Modular Microservices Architecture

- The system must support independent deployment of microservices.

2) NFR-MAIN-002: Logging & Monitoring

- Integrated log monitoring, performance alerts, API health status dashboards.

3) NFR-MAIN-003: Version Control

- All application updates must follow semantic versioning and maintain backward compatibility for existing data.



6.8. Data Quality & Integrity Requirements

- 1) **NFR-DATA-001: Data Accuracy**
 - Automated duplicate detection must prevent multiple volunteer profiles.
- 2) **NFR-DATA-002: Integrity Checks**
 - Mandatory fields (e.g., MyBharat ID, Aadhaar, DOB) must be validated for format, completeness, and correctness.
- 3) **NFR-DATA-003: Evidence Integrity**
 - Every evidence upload must store:
 - Timestamp
 - Geo-location
 - Nodal Officer ID
 - Hash signature for tamper detection
- 4) **NFR-DATA-004: Certificate Integrity**
 - Certificates must be digitally signed (new scheme) and tamper-proof.

6.9. Reporting & Analytics Requirements

- 1) **NFR-REP-001: Real-Time Sync**
 - Dashboards should reflect near real-time data (< 5-minute delay).
- 2) **NFR-REP-002: Export Capability**
 - All MIS views must support PDF, Excel, and CSV exports.
- 3) **NFR-REP-003: Predictive Analytics**
 - System must support machine learning models for readiness scoring, dropout prediction, and risk heatmaps.
- 4) **NFR-REP-004: Historical Data**
 - Store historical records for at least **2 years** for audit and national reporting.

6.10. Compliance Requirements

- 1) **NFR-COMP-001: Government of India Standards**
 - Comply with MeitY guidelines for hosting, API security, data protection, and accessibility.
- 2) **NFR-COMP-002: Hosting Environment**
 - System must be deployed on **Government Community Cloud (GCC)** or MeitY-approved infrastructure.
- 3) **NFR-COMP-003: Logging Standards**
 - All logs must follow CERT-In guidelines for retention and review.

7. Business Rules / Business Logic

Business Rules define the mandatory logic, constraints, validations, eligibility conditions, and operational constraints that govern how the Unified Aapda Mitra Platform must function. These rules ensure data integrity, process standardization, compliance adherence, and consistent decision-making across the NDMA ecosystem.

7.1. Registration & Eligibility Rules

1) **BR-REG-001: Mandatory 30-Field Profile**

- A volunteer record shall not be considered valid until **all 30 mandatory fields** (as defined by NDMA) are completed by the DDMA Nodal Officer.

2) **BR-REG-002: Mandatory MyBharat ID**

- Every volunteer record must include a **valid MyBharat ID**.
For old scheme volunteers, this must be collected during the next profile verification cycle.

3) **BR-REG-003: Aadhaar-Based Identity Validation**

- Aadhaar authentication is mandatory for all YAMS volunteers and optional-but-recommended for UAMS volunteers.

4) **BR-REG-004: Duplicate Prevention Rule**

A volunteer cannot be registered more than once. Duplicate detection is enforced using:

- Aadhaar number
- Mobile number
- MyBharat ID
- Youth Organisation ID (if applicable)

5) **BR-REG-005: Youth Organisation Affiliation Rule**

- A volunteer submitted under YAMS must be validated by the respective Youth Organisation (NCC, NSS, NYKS, BS&G, MyBharat, NCSC).
Unverified affiliations must be flagged for DDMA review.

7.2. Verification Rules

1) **BR-VER-001: Auto-Eligibility Rule**

- Age limits
- Location mapping
- Youth organisation affiliation validation
- Scheme applicability (UAMS/YAMS)

2) **BR-VER-002: Incomplete Profiles Cannot Proceed**

- Volunteers with incomplete profiles or mismatches must remain in **“Pending Verification”** status until resolved.

3) **BR-VER-003: Verification Authority Rule**

- Only DDMA Nodal Officers may perform verification; SDMA/National level users cannot modify profile attributes.

7.3. Training Rules

1) **BR-TRN-001: Batch Allocation Rule**

- A volunteer cannot be assigned to more than one training batch at a time.

2) **BR-TRN-002: Attendance Eligibility Rule**

- Volunteers must attend a minimum NDMA-defined percentage (e.g., **90%**) to qualify for certificate issuance.

3) **BR-TRN-003: Attendance Capture Authority Rule**

- Only **DDMA Nodal Officers** may capture attendance and upload evidence.
Training Institutes do *not* enter data into the MIS.

4) **BR-TRN-004: Geo-Tagging Rule**

Every attendance evidence upload (photo/video) must include:

- GPS location
- Timestamp
- Device/Nodal Officer ID

5) **BR-TRN-005: Evidence Approval Rule**

- Training evidence must be reviewed and approved by the **NDMA Super Admin** before the training batch can be marked “Completed.”

6) **BR-TRN-006: Offline Attendance Rule**

- If offline attendance is captured, it must sync automatically when the device returns online.

7) **BR-TRN-007: Cross-Scheme Training Rule**

- A volunteer can undergo only **one training cycle** under the scheme they belong to (UAMS or YAMS)

7.4. Certification Rules

1) **BR-CERT-001: Certificate Issuance Condition**

A certificate may be issued only if:

- Training completed
- Attendance % met
- Evidence approved
- Volunteer meets NDMA criteria

2) **BR-CERT-002: Old Scheme Certificate Rule (UAMS)**

- Old scheme volunteers must receive **physical certificates only** unless migration is completed.

3) **BR-CERT-003: New Scheme Certificate Rule (YAMS)**

- New scheme certificates must be issued digitally via the **DigiLocker API**.

4) **BR-CERT-004: Certificate Re-Issue Rule**

- Certificates may be re-issued **ONLY** with NDMA approval and complete audit logging.

7.5. ERK Kit & Insurance Rules

1) **BR-ERK-001: ERK Distribution Dependency Rule**

ERK kits shall be marked distributed only after:

- Volunteer completes training
 - Certificate eligibility criteria met
- ### 2) **BR-ERK-002: Insurance Activation Rule**
- Insurance coverage shall begin **only after** NDMA-approved training completion.
- ### 3) **BR-ERK-003: Acknowledgment Rule**
- Volunteers must acknowledge ERK & insurance receipt via mobile app.
- ### 4) **BR-ERK-004: ERK/Insurance Feedback Rule**
- Feedback shall be collected as part of the **Unified Feedback Form**.

7.6. Deployment & Geo-Tracking Rules

1) **BR-DEP-001: Deployment Assignment Rule**

- Only NDMA/SDMA/DDMA users may assign deployment tasks.

2) **BR-DEP-002: Sachet Geo-Tracking Rule**

For new YAMS volunteers:

- Live tracking must be retrieved through Sachet APIs
- Location updates are continuous during deployment

3) **BR-DEP-003: Legacy Location Rule**

For old UAMS volunteers:

- Location updates must be uploaded as Excel sheets until transitioned to API-based model
- ### 4) **BR-DEP-004: Non-Compliance Deployment Rule**
- Volunteers failing monthly reporting or readiness updates may be marked “Inactive” for future deployments.
- ### 5) **BR-DEP-005: Emergency Escalation Rule**
- During national emergencies, NDMA may override state-level deployment rules.

7.7. Monthly Reporting Rules

1) **BR-MON-001: Monthly Reporting Mandate**

- All volunteers must submit a monthly readiness report.

2) **BR-MON-002: Delinquency Rule**

- If no update is submitted for **3 consecutive months**, the volunteer must be flagged as “Delinquent.”

3) **BR-MON-003: Escalation Rule**

Delinquency must escalate automatically to:

- DDMA (first alert)
- SDMA (second alert)
- NDMA (third alert)

7.8. Feedback Rules

1) **BR-FDB-001: Unified Feedback Collection Rule**

- Training, ERK, insurance, and overall experience feedback must be captured in a single unified form.

2) **BR-FDB-002: Mandatory Feedback Rule**

- Feedback is mandatory at the end of training.

3) **BR-FDB-003: Feedback Analytics Rule**

- Feedback must feed into state and national dashboards for quality improvement.

7.9. Dashboard Rules

1) **BR-DASH-001: Public Dashboard Visibility Rule**

- The public dashboard must display **only aggregated, non-PII data**.

2) **BR-DASH-002: MIS Role-Based Rule**

- NDMA, SDMA, DDMA dashboards must show data based on jurisdiction access only.

3) **BR-DASH-003: Real-Time Data Rule**

- Dashboards must refresh every 5 minutes or support manual refresh.

7.9. Data Integrity & Governance Rules

1) **BR-DATA-001: Immutable Evidence Rule**

- Attendance evidence cannot be deleted once uploaded.

2) **BR-DATA-002: Audit Logging Rule**

- Every key action must be logged with timestamp, user ID, and action metadata.

3) **BR-DATA-003: Historical Preservation Rule**

- Volunteer history must be retained for a minimum **2 years**.

4) **BR-DATA-004: State Jurisdiction Rule**

- States must only view and manage volunteers belonging to their own districts.

8. Portal Work Flow

The Unified Aapda Mitra Platform must support a set of standardized workflows that span the complete lifecycle of a volunteer under both UAMS (Old Scheme) and YAMS (New Scheme). Each workflow describes the sequence of actions performed by system actors, integrated services, and automation components.

The workflows below reflect the **latest NDMA instructions**, including:

- 30-field volunteer profiling

- Mandatory MyBharat ID
- Auto-eligibility determination
- Evidence capture by DDMA
- NDMA-level evidence approval
- DigiLocker integration (new scheme)
- Sachet App integration
- Youth Organisation API intake
- Dashboard updates (public + MIS)

8.1. Workflow 1 — Volunteer Registration & Onboarding

Actors: Volunteer, Youth Organisation, DDMA, Aadhaar Service, System

Process Steps

1. **Volunteers view scheme information** on the public Aapda Mitra website/mobile app.
2. **Volunteer initiates registration requests** or provides consent for youth-organisation–driven registration.
3. **Youth Organisation submits a volunteer list** via secure API/upload (for YAMS).
4. **System pre-validates records** for duplicates (Aadhaar + Mobile + MyBharat ID).
5. **DDMA Nodal Officer completes the 30-field profile**, ensuring MyBharat ID is captured.
6. **Aadhaar authentication service validates identity**.
7. **Auto-eligibility engine evaluates rules** based on NDMA criteria.
8. System updates status to **“Verified & Eligible”** or **“Pending for Clarification”**.
9. **Notifications sent** via SMS/Email/WhatsApp.
10. Dashboard updates for DDMA/SDMA/NDMA.

8.2. Workflow 2 — Youth Organisation Affiliation

Actors: NCC/NSS/NYKS/BS&G/MyBharat/NCSC, DDMA, System

Process Steps

1. Youth organisation uploads volunteer dataset via API or file.
2. System validates data structure and mandatory fields.
3. Youth organisation confirms affiliation for each volunteer.
4. System sends flagged entries (mismatches) to DDMA for verification.
5. Verified records move to auto-eligibility stage.
6. Dashboard reflects validated vs pending youth datasets.

8.3. Workflow 3 — Volunteer Verification & Approval

Actors: DDMA, SDMA (review), Aadhaar API, System

Process Steps

1. DDMA reviews volunteer details (demographics, organisation, supporting documents).
2. Aadhaar validation confirms identity.
3. System applies **auto-eligibility rules** (scheme-specific criteria).
4. SDMA receives a view-only summary of eligible volunteers.
5. Workflow proceeds automatically to Training Allocation Module.
6. Audit logs capture all verification actions.

8.4. Workflow 4 — Training Allocation

Actors: DDMA, SDMA, System

Process Steps

1. SDMA onboards and configures training institutes.
2. DDMA creates training batches and maps volunteers.
3. System checks batch capacity constraints.
4. Volunteers receive automated notifications (SMS/Email/WhatsApp).
5. Institute dashboard displays batch rosters.
6. Dashboard metrics updated: "Volunteers Allocated", "Batches Scheduled".

8.5. Workflow 5 — Attendance & Evidence Capture

Actors: DDMA Nodal Officer, NDMA Super Admin, System

Process Steps

1. Nodal officer initiates attendance marking each training day.
2. System generates QR code for validation.
3. Officer captures **geo-tagged photo/video evidence**.
4. Evidence uploaded into centralized repository.
5. System verifies completeness (GPS + timestamp + officer ID).
6. NDMA Super Admin reviews and **approves** or **rejects** evidence.
7. If rejected, DDMA re-uploads corrected evidence.
8. Attendance is auto-calculated for certificate eligibility.

8.6. Workflow 6 — Certification (Physical + DigiLocker)

Actors: System, NDMA, Volunteer

Old Scheme (UAMS)

1. System verifies training completion + attendance + evidence approval.
2. Certificates are generated physically by district/state.
3. System updates certificate status in MIS.



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New Scheme (YAMS)

1. System pushes eligible data to DigiLocker API.
2. Volunteer receives certificate via DigiLocker wallet.
3. MIS marks certificate as issued.

8.7.Workflow 7 — ERK Kit & Insurance Issuance

Actors: SDMA, DDMA, Volunteer, System

Process Steps

1. SDMA updates ERK logistics (distribution records, inventory).
2. DDMA issues ERK kit to each eligible volunteer.
3. Volunteer acknowledges receipt on mobile app.
4. SDMA activates insurance policy (3-year validity).
5. System updates ERK/Insurance dashboard metrics.

8.8.Workflow 8 — Deployment & Field Operations

Actors: NDMA, SDMA, DDMA, Volunteer, Sachet App

Process Steps

1. NDMA/SDMA/DDMA creates deployment tasks.
2. Volunteers receive notifications.
3. For YAMS volunteers → Sachet app sends **real-time location data** to the MIS.
4. For UAMS volunteers → DDMA uploads Excel-based location mappings.
5. Volunteers update field progress (status, observations).
6. NDMA views national deployment heatmaps.

Actors: SDMA, DDMA, Volunteer, System

Process Steps

1. SDMA updates ERK logistics (distribution records, inventory).
2. DDMA issues ERK kit to each eligible volunteer.
3. Volunteer acknowledges receipt on mobile app.
4. SDMA activates insurance policy (3-year validity).
5. System updates ERK/Insurance dashboard metrics.



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8.9.Workflow 9 — Monthly Reporting & Compliance

Actors: Volunteer, DDMA, SDMA, System

Process Steps

1. Volunteer submits monthly readiness report.
2. System triggers auto-reminders for non-compliance.
3. Escalations sent to DDMA → SDMA → NDMA.
4. MIS dashboards reflect compliance & delinquency metrics.

8.10.Workflow 10 — Feedback Management

Actors: Volunteer, System, Analytics Engine

Process Steps

1. Volunteer submits feedback on:
 - Training
 - ERK kit
 - Insurance
 - Overall experience
2. The system categorizes feedback by theme.
3. Feedback flows into analytics dashboards.
4. SDMA & NDMA monitor trends and take corrective action.

8.11.Workflow11-Public Dashboard Updates

Actors: Public, System, NDMA

Process Steps

1. MIS aggregates non-PII national data.
2. Public dashboard displays metrics: total volunteers, districts covered, batches, youth organisation contributions, etc.
3. Data auto-refreshes every 5 minutes.
4. NDMA controls what metrics appear publicly.

8.12.Workflow 12 — MIS Dashboards (DDMA, SDMA, NDMA)

Actors: District, State, National Officials

Process Steps



1. MIS aggregates role-based data.
2. DDMA views training, attendance, evidence status.
3. SDMA views state readiness, ERK & insurance data, performance metrics.
4. NDMA views national readiness index, deployment heatmaps, compliance levels.
5. Predictive analytics highlight risk zones and training gaps.

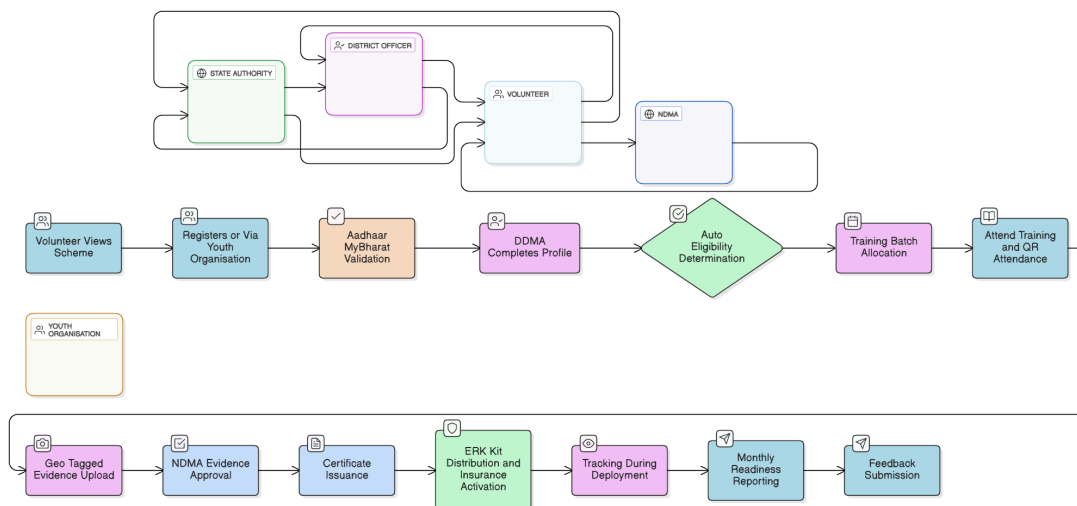


Fig1: User Journey

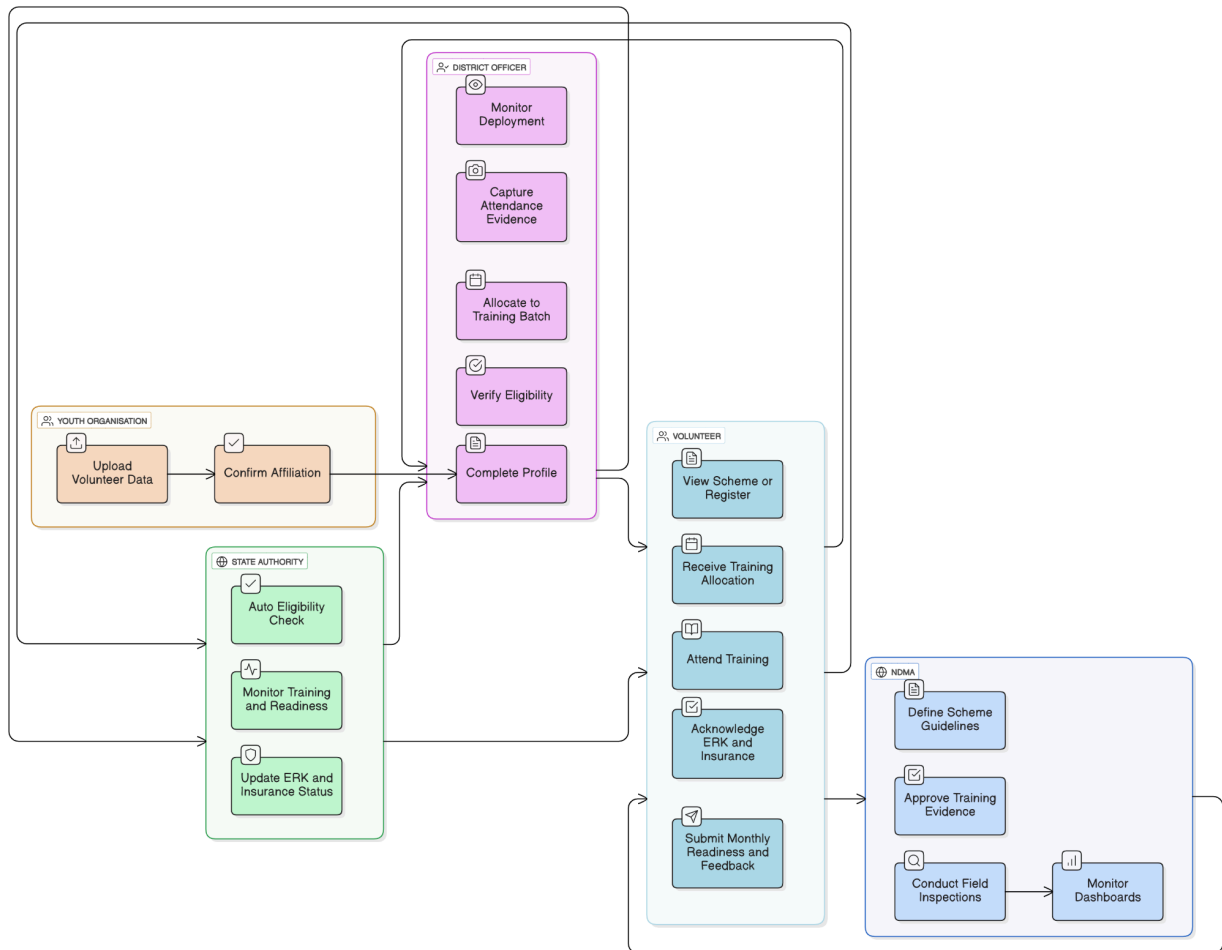


Fig2: Swimlane Diagram

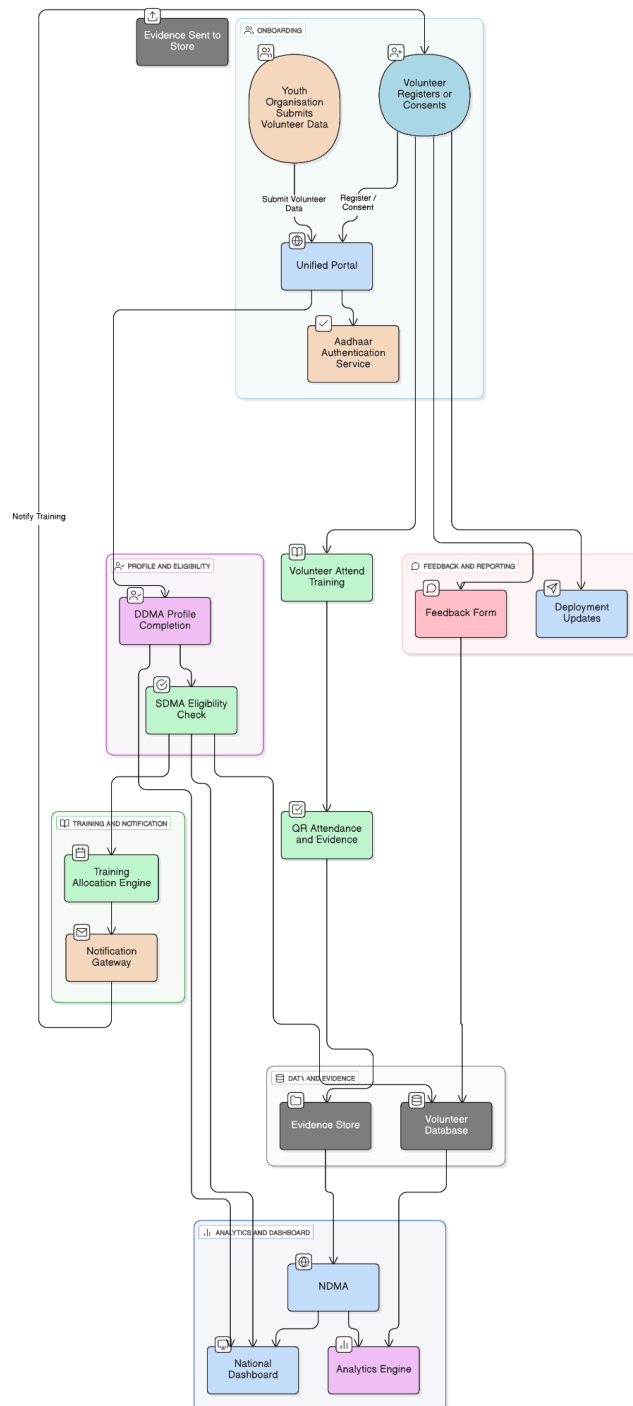


Fig2: Data Fl



12. Key Abbreviation

Abbreviation	Full Form / Meaning
NDMA	National Disaster Management Authority
SDMA	State Disaster Management Authority
DDMA	District Disaster Management Authority
UAMS	Upscaling of Aapda Mitra Scheme (Old Scheme)
YAMS	Yuva Aapda Mitra Scheme (New Scheme)
UMAS	Upscaling of Mitra Aapda Scheme (legacy reference in old docs)
NDRF	National Disaster Response Force
SDRF	State Disaster Response Force
ERK	Emergency Response Kit
MoYAS	Ministry of Youth Affairs and Sports (parent body for youth orgs)
MyBharat	My Young India Platform (National Youth Identity used in YAMS)
NCC	National Cadet Corps
NSS	National Service Scheme
NYKS	Nehru Yuva Kendra Sangathan
BS&G	Bharat Scouts and Guides
NCSC	National Cadet Special Category / Youth Group under YAMS
TOT	Training of Trainers (Master Trainer Programme)
DRR	Disaster Risk Reduction
PPE	Personal Protective Equipment (part of ERK)

******End of document******